

Lab task 2

The objectives of the question are to:

- Implement the constructor, accessors and methods
- understand object composition
- write test driver to test the methods

You are required to write different class in separate files. **Your submission is expected to reflect this.**

Examine the following class specifications, additional information and answer the questions that follow:

```
class Date
{Private members :
    int dd;
    int mm;
    int yy;
public members:
    Date(int = 1, int=1,int=2017);
    void setDate(int,int, int);
    int getDD;//property
    int getMM; //property
    int getYY; // property
}

class Appointment
{
Private members
    string NRIC; //readonly
    string name;
    double fees;
    Date visit;
    Date nextVisit;
public members
    Appointment(string, string, double, Date,Date);
    void setAppt(string, double, Date,Date);
    String getVisit; //property
    String getnextVisit //property
    String getName
    void displayAppt();
    void displayAppt(string);
}
```

Additional Information:

Method	Remarks
Class: Date	
Date(int = 1, int=1, int=2015)	Constructor with default values.
void setDate(int, int, int)	Method Set data members of Date class to the pass-in values accordingly.
int getDD int getMM int getYY	Property. Return the data members of Date class respectively.
Class: Appointment	
Appointment(string, string, double, Date,Date)	Default Constructor of Appointment class. Set data

	members, <i>NRIC</i> and <i>name</i> to null, <i>fees</i> to 0.0
<code>void setAppt(string, double, Date, Date)</code>	Function Set the data members to the pass-in arguments' values accordingly.
<code>String getVisit</code> <code>String getNextVisit</code> <code>String getName</code>	Property. Return the data members, <i>visit</i> / <i>nextVisit</i> / <i>name</i> as string respectively.
<code>void displayAppt()</code>	Use <code>Console.WriteLine</code> to display the data members of Appointment class. If the data member of <i>Date</i> class is 0, display a dash '-'.
<code>void displayAppt(string)</code>	Use <i>console write function</i> to display the <i>name</i> and <i>fees</i> of Appointment class when the <i>NRIC</i> data member of the instance matches the pass-in value. Invokes <i>displayAppt()</i> ;

Note¹: When there is no next visit, the data members, *dd*, *mm* and *yy* in *Date* class are set to 0.

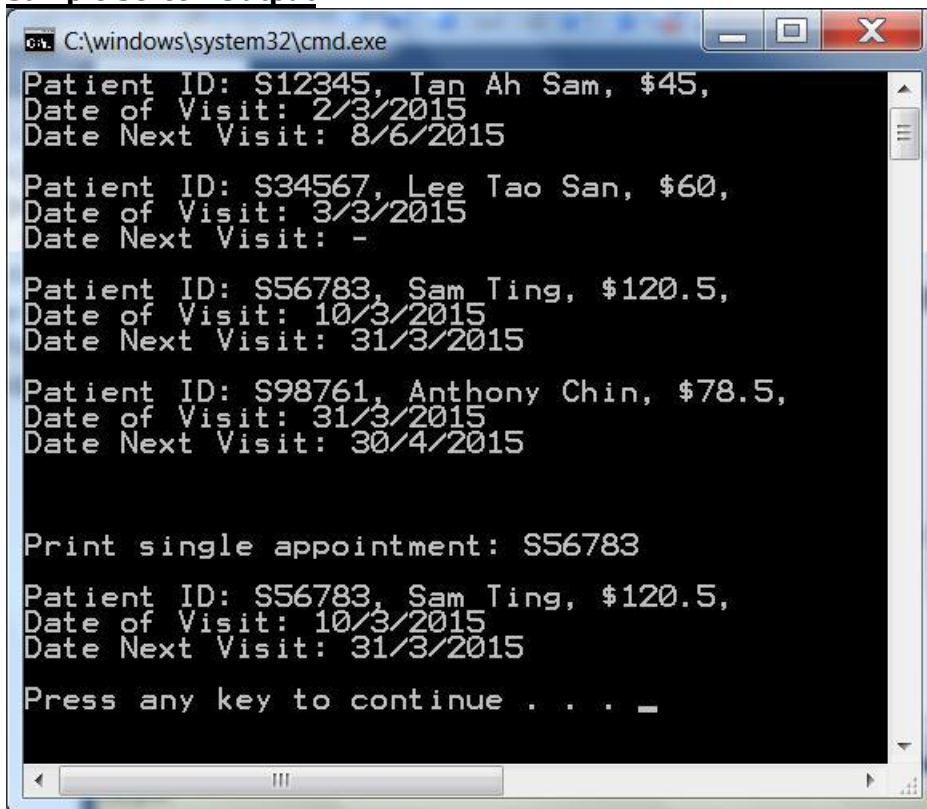
- (b) Develop C# code to implement **ALL** the methods found in the above-mentioned class specifications:
- (i) Date Class
 - (ii) Appointment Class
- (c) Write a test driver to test **ALL** the methods in the Date and Appointment classes. The test driver should include the following:
- (i) declare 4 Appointment object
 - (ii) initialised the Appointment objects with the data from the test data appended below:
 - (iii) display all appointment objects, using the default *displayAppt()* method
 - (iv) display a single appointment object, using *displayAppt(string)* method, passing in the NRIC value, "S56783".

Include a screen output similar to the sample screen output shown below in your submission.

Test Data

NRIC	Name	Fees	Date of Visit	Date of next visit
S12345	Tan Ah Sam	45.00	2 March 2015	8 June 2015
S34567	Lee Tao San	60.00	3 March 2015	-
S56783	Sam Ting	120.50	10 March 2015	31 March 2015
S98761	Anthony Chin	78.50	31 March 2015	30 April 2015

Sample Screen Output



A screenshot of a Windows command prompt window. The title bar shows the file name 'G:\ C:\windows\system32\cmd.exe'. The window has standard Windows window controls (minimize, maximize, close) on the right. The command prompt displays several lines of text representing patient appointment data. The text is as follows:

```
Patient ID: S12345, Tan Ah Sam, $45,  
Date of Visit: 2/3/2015  
Date Next Visit: 8/6/2015  
  
Patient ID: S34567, Lee Tao San, $60,  
Date of Visit: 3/3/2015  
Date Next Visit: -  
  
Patient ID: S56783, Sam Ting, $120.5,  
Date of Visit: 10/3/2015  
Date Next Visit: 31/3/2015  
  
Patient ID: S98761, Anthony Chin, $78.5,  
Date of Visit: 31/3/2015  
Date Next Visit: 30/4/2015  
  
Print single appointment: S56783  
  
Patient ID: S56783, Sam Ting, $120.5,  
Date of Visit: 10/3/2015  
Date Next Visit: 31/3/2015  
  
Press any key to continue . . . _
```

The command prompt has a scrollbar on the right side and a status bar at the bottom.